

SANDEEP POLISETTY

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EDUCATION

University of Massachusetts Amherst

Ph.D., Computer Science

Aug 2020 – Aug 2025(Expected)

Amherst, MA

University of Massachusetts Amherst

M.S, Computer Science

Aug 2016 – May 2020

Amherst, MA

Indian Institute of Technology

B.Tech, M.Tech, Naval Architecture

Sep 2008 – May 2013

Kharagpur, India

GOALS

I aspire to build infrastructure that is fueling the AI revolution. Efficient and easy to adopt frameworks are especially important with rising hardware costs. To support these goals, I have worked at various levels across the framework such as efficient kernel design, parallelization strategies, compiler abstractions and hardware aware models to optimize performance.

SELECTED PUBLICATIONS

GSplit: Scaling graph neural network training on large graphs via split-parallelism

S Polisetty, J Liu, K Falus, YR Fung, SH Lim, H Guan, M Serafini

MLsys

2025

Accelerating graph sampling for graph machine learning using GPUs

A Jangda, S Polisetty, A Guha, M Serafini

Eurosys

2021

Graphmini: Accelerating graph pattern matching using auxiliary graphs

J Liu, S Polisetty, H Guan, M Serafini

PACT

2023

WORK EXPERIENCE

Intel

AI Frameworks Engineer Intern

May 2024 – August 2024

Santa Clara, CA

- Worked on the torch extensions for large language models using inductor framework to utilize hardware optimized frameworks such as OneDNN and XeTLA
- Benchmarked the performance of quantization optimizations on important families of GenAI models such as Llama and Qwen

Reservoir Labs (acquired by Qualcomm)

Research Engineer Intern

May 2021 – August 2021

Remote, NY

- Extended TVM intermediate representation Relay to support dynamic shape operators (eg. embedding bag) and generate c-code for downstream polyhedral compiler.
- Wrote transformation passes such as shape propagation and affine loop identification to increase the scope for polyhedral optimizations which are restricted to affine loops

Amazon

Software Engineer Intern

May 2018 – August 2018

Seattle, WA

- Extended in-house consistency service, to support zookeeper clients while maintaining original consistency protocol.
- Documented proof of correctness and unit test pass rate leading to improved adoption by end users.

Healthkart.com

Software Engineer

May 2019 - May 2021

Gurgaon, India

- Worked on building scalable Recommender systems for Indian e-commerce platform.

RESEARCH EXPERIENCE

Distributed graph machine learning

advised by Prof. Marco Serafini and Prof. Hui Guan

January 2021 - Present

- Designed a novel distributed sampling and training paradigm to eliminate redundancy across multiple mini-batches in GNN training across multiple GPUs
- Designed efficient abstractions to minimally change existing training and sampling loops. Light weight novel data structures and optimal CUDA kernels were designed to perform inter-GPU communication and co-ordination efficiently.
- Performed extensive experiments across different model architectures and hardware configurations. Our approach allows for deeper GNN models and outperforms the state of the art benchmarks such as Quiver and DGL upto **2x**. (work under submission)

Large Scale Semantic Scholar Paper Recommendation

at John Hopkins Speech and Language Workshop

June 2023 - Aug 2023

- Processed massive semantic scholar database consisting of all published research, utilizing highly parallel cluster computing to create datasets consumable by deep learning frameworks.
- Used temporal partitioning of the graph and unified virtual addressing to train an extremely large graph containing over 500 GB of data on commodity V100 GPUs.
- Decreased training time further by using edge selection to limit training edges. The resulting GAT model delivered competitive accuracy relative to state of the art techniques based on matrix factorization, as well as decreasing training time to under a day from one week. (currently under submission)

GPU sampling for GNN

advised by Prof. Marco Serafini

September 2020 - March 2021

- Integrated a novel in-GPU transit node-centric sampler called Nextdoor across various state-of-the-art graph training pipelines using torch extension to demonstrate the robustness of the designed abstraction and end to end training performance improvements (Published at Eurosys - 2021)

Efficient Graph Mining

advised by Prof. Marco Serafini

November 2020

- Worked on eliminating redundancy in graph pattern mining by constructing light weight intermediate structures. Modified existing graph pattern matching algorithms to utilize these light weight structures resulting in speedups upto **2x** over the state of the art frameworks. (Published at PACT-2023)

TECHNICAL SKILLS

Languages: C/C++, CUDA, python, java

Frameworks: tensorflow, pytorch, DGL, TVM, pybind, NSightSystems

SERVICES

- Reviewer Pattern Analysis and Machine Intelligence
- Reviewer SC-2024
- Artifact Evaluator OSDI-2024, ATC-2024
- Artifact Evaluator OSDI-2020